

MACHINE LEARNING

CLASSIFICATION FRAMEWORK

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Classification: introduction

- Classification, also called **pattern recognition**, aims to predict **qualitative** response variables Y that take values in an unordered set $\mathcal{G} = \{\mathcal{G}_1, \dots, \mathcal{G}_q\}$, e.g.

`infected` $\in \{\text{Yes}, \text{No}\}$

`eye color` $\in \{\text{brown}, \text{blue}, \text{green}\}$

- Given the vector of predictors $X = (X_1, \dots, X_p)$, the classification task is to build a function G , called **classifier**, that takes the vector X and predicts the value of Y , i.e., $G(X) \in \mathcal{G}$.

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- ▶ Often one is more interested in the **probability** that Y belongs to each class $\mathcal{G}_1, \dots, \mathcal{G}_q$, e.g.,
 - ▶ the probability that a client of a bank will default;
 - ▶ the probability that a cancer is malignant;
 - ▶ etc.
- ▶ We typically identify the q classes with the numbers 1 to q , such that $\mathcal{G} = \{1, \dots, q\}$, or if we only have two classes (**binary data**), then $\mathcal{G} = \{0, 1\}$.

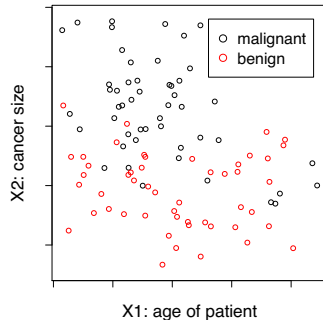
Classification: introduction

Example.: [Classification of cancer; $p = 2$, $q = 2$]

- We want to find a classifier

$$G(\text{age}, \text{size}) \in \mathcal{G} = \{\text{benign}, \text{malignant}\},$$

that predicts whether a cancer is **benign** or **malignant**, based on the predictors **age** and **size**.



The 0–1 loss function

- ▶ How do we measure the quality of prediction?
- ▶ For classification, squared error loss might not be optimal. For a prediction $\hat{y}$ we usually use the **0 – 1 loss function**

$$L(y, \hat{y}) = \mathbf{1}\{y \neq \hat{y}\} = \begin{cases} 0 & \text{if } y = \hat{y} \\ 1 & \text{if } y \neq \hat{y} \end{cases}$$

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- ▶ For a classifier $\hat{G}$, the **expected misclassification rate** is

$$\text{Err}_{\hat{G}} = \mathbb{E}[\mathbf{1}\{Y \neq \hat{G}(X)\}] = \Pr\{Y \neq \hat{G}(X)\},$$

where the expectation is taken over (X, Y) .

- ▶ Note: Also other loss functions for classification tasks are common, since
 - ▶ in medicine, for example, false negative and false positive are not equally severe, and we want to penalize one type of error more than the other;
 - ▶ the 0-1 loss is not smooth and may lead to optimization problems.

The Bayes classifier

- ▶ What is a good **classifier**? Ideally, we would like to find the optimal G^* that **minimizes** $\text{Err}_{\hat{G}} = \Pr\{Y \neq \hat{G}(X)\}$.
- ▶ For simplicity, consider two-class classification, i.e. $\mathcal{G} = \{0, 1\}$.
- ▶ It can be shown, that the optimal G^* is the so-called **Bayes classifier**, which assigns to a predictor x the **most likely class**, i.e.,

$$G^*(x) = \operatorname{argmax}_{j=0,1} \Pr(Y = j \mid X = x) = \begin{cases} 0 & \text{if } \Pr(Y = 0 \mid X = x) > 1/2 \\ 1 & \text{otherwise.} \end{cases}$$

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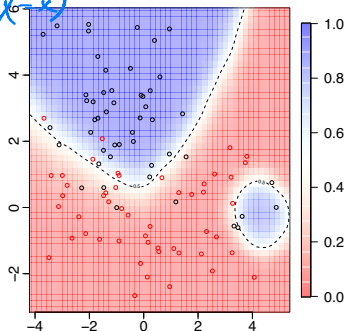
$\hat{P}(Y=j|X=x)$

- ▶ The **Bayes decision boundary** is the curve of all $x \in \mathbb{R}^p$ with

$$\Pr(Y = 0 \mid X = x) = 1/2.$$

- ▶ The Bayes classifier gives the smallest possible 0 – 1 error. It is the irreducible error, also called the **Bayes error rate**:

$$\text{Err}_{G^*} = \Pr\{Y \neq G^*(X)\}.$$



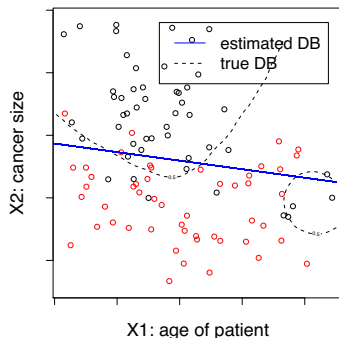
The Bayes decision rule

- ▶ The Bayes classifier motivates to estimate the **posterior class probabilities** $\widehat{\Pr}(Y = j \mid X = x)$, $j = 0, 1$, and then use the **plug-in decision rule**

$$\widehat{G}(x) = \begin{cases} 0 & \text{if } \widehat{\Pr}(Y = 0 \mid X = x) > 1/2 \\ 1 & \text{otherwise.} \end{cases}$$

- ▶ The **decision boundary** is the line where

$$\widehat{\Pr}(Y = 0 \mid X = x) = \widehat{\Pr}(Y = 1 \mid X = x) = 1/2.$$



Statistical modeling of $\hat{G}$

- (a) **Generative models:** estimate the joint distribution of (Y, X) and deduce the posterior class probabilities for $j = 0, 1$

$$\widehat{\Pr}(Y = j \mid X = x) = \frac{\widehat{\Pr}(Y = j, X = x)}{\widehat{\Pr}(X = x)}.$$

- ▶ This is often more complex and needs more data.
- ▶ Examples: **Linear/Quadratic Discriminant analysis**, **Naive Bayes**, etc.

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- (b) **Discriminative models:** estimate only the conditional distribution $(Y \mid X)$, that is, directly the posterior class probabilities $\widehat{\Pr}(Y = j \mid X = x)$.

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- (c) **Geometric models:** directly estimate a function that predicts the class 0, 1 of y from x .

- ▶ Simple, since no distribution is needed, but does not provide estimates for the probabilities of the class predictions.
- ▶ Examples: **Support vector machines**

The empirical misclassification rate

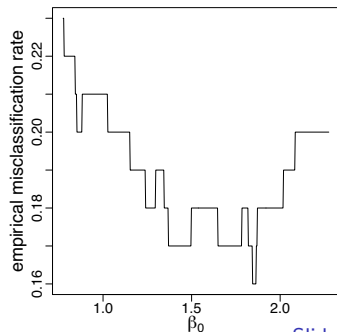
Given a **training data** set $(x_1, y_1), \dots, (x_n, y_n)$, with $x_i \in \mathbb{R}^p$ and $y_i \in \{1, \dots, q\}$, how can we estimate the model $\hat{G}$?

- ▶ In regression, we minimize the empirical mean squared error on the training set. We could do the same for the **empirical misclassification rate** by minimizing

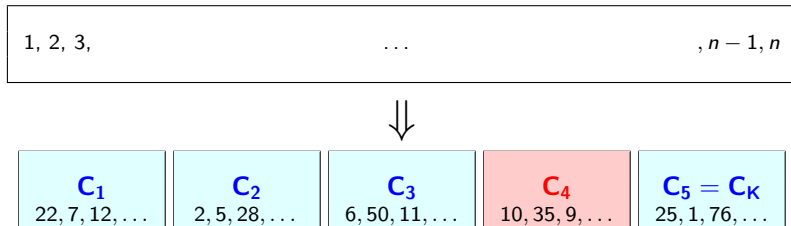
$$\text{err}_{\hat{G}} = \frac{1}{n} \sum_{i=1}^n \mathbf{1}\{y_i \neq \hat{G}(x_i)\}$$

- ▶ Problem: The 0-1 loss function $\mathbf{1}\{y \neq \hat{y}\}$ is discontinuous, which makes **numerical optimization** very hard.

- ▶ Solution: Different methods such as **maximum likelihood** are used for **parameter estimation**, which are close to 0 – 1 loss but differentiable.
- ▶ The 0 – 1 loss can still be used for **model selection**, or to compare models fitted with different methods.



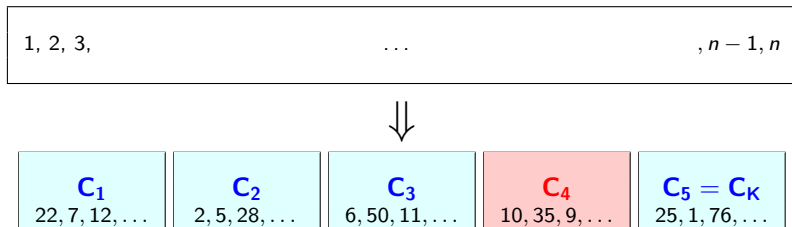
K-Fold Cross-Validation



- ▶ For a data set $\{(x_1, y_1), \dots, (x_n, y_n)\}$ we perform **K-Fold Cross-Validation** as in the regression framework.
- ▶ For each fold C_k , $k = 1, \dots, K$ we treat C_k as **validation/test** set and fit a classification model $\hat{G}_{-k}$ on the remaining $K - 1$ folds. We then compute the empirical misclassification rate

$$\text{err}_k = \frac{1}{|C_k|} \sum_{i \in C_k} L(y_i, \hat{G}_{-k}(x_i)) = \frac{1}{n/K} \sum_{i \in C_k} \mathbf{1}\{y_i \neq \hat{G}_{-k}(x_i)\}.$$

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- ▶ The **K-fold CV estimate** is the average of these values

$$\text{CV}_{(K)} = \frac{1}{K} \sum_{k=1}^K \text{err}_k.$$